Refeeding after fasting in the rat: effects of carbohydrate[1-3]

Per Björntorp, Mei-Uih Yang, and MRC Greenwood

ABSTRACT In order to find ways to modify adaptations during refeeding after fasting the amount of carbohydrate fed during refeeding was varied, or its absorption slowed by a glucosidase inhibitor. Exchanging dietary carbohydrate with fat diminished refeeding adaptations in carbohydrate pathways (glycogen contents of liver and muscle, glucose incorporation into carbondioxide, and triglyceride in adipose tissue), but instead facilitated lipid storage (weight of and lipoprotein lipase activity in adipose tissue). An early specific effect of carbohydrate refeeding was an increase in the insulin response in glucose metabolic pathways in adipose tissue. Inhibiting carbohydrate absorption by a glucosidase inhibitor (Acarbose) caused a delay in energy accumulation during refeeding both in glycogen stores, and, more markedly, in adipose tissue triglyceride stores. These data are consistent with the interpretation that the drug prevents or delays carbohydrate uptake, and causes less adaptations of adipose tissue metabolic pathways for lipid storage. The tissue adaptations during refeeding after fasting seem to depend on the energy substrate reaching these tissues. *Am J Clin Nutr* 1983;37:396–402.

KEY WORDS Fasting, refeeding, glycogen, lipid, carbohydrate, fat, glucosidase inhibitor, Acarbose

Introduction

The effects of refeeding after fasting on tissue metabolism are known in great detail (for review see References 1 and 2). Recently the role of these factors in overall organism energy balance have become of interest in developing an understanding of the failures of weight maintenance after reduction. In an attempt to find explanations to the rapid relapse after treatment of obesity, a model has been used where rats have been refed after a brief fast and adaptations in energy substrate fluxes in relation to tissue energy metabolism followed. The main findings have been increased fluxes of energy from the circulation into tissues effectively adapted to catch energy. This has the consequence that concentrations of circulating energy substrate are lower, producing for example hypoglycemia (2–4). Furthermore, gut enzyme activities, weight, and cellularity also adapt to facilitate energy capture (5). In addition, the effectiveness of energy capture in protein, carbohydrate, and fat stores (food efficiency) is greatly enhanced in the nonobese, starved rat (3), but this is not necessarily the case in dietary obese rats (6).

It is known from previous work that refeeding diets rich in carbohydrate produces pronounced enzyme adaptations in liver and adipose tissue (see reviews cited above). This is also evident from previous work in this series in which glucose metabolic pathways have been markedly adapted when studied in vitro in adipose tissue (2). Furthermore, synthesis of glycogen and fatty acids in liver and adipose tissues exceed control values when studied in vivo (4), and flux of glucose from circulation into the adapted tissues is markedly increased (4). All these observations made it logical to study specifically the question whether modification of the amount of carbohydrate reaching the energy-storing tis-

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sues would alter the adaptive response during the refeeding phase. Therefore the following study was designed to assess the impact of *1*) decreasing the carbohydrate content by adding fat to the refeeding diet, *2*) slowing carbohydrate absorption by giving an intestinal glucosidase inhibitor or, *3*) varying carbohydrate contents of the refeeding diet with additional sucrose. A preliminary report of some of these results has been given (7).

Experimental procedures

Study I. Fat feeding

This study was designed and performed as previously described in detail (3, 4). Briefly, male Sprague-Dawley rats weighing 200 to 250 g were starved for 3 nights and 2 days (65 h), and then refed up to 8 days ad libitum with either Purina Rat Chow (22.8, 4.5, and 52.9% of protein, fat, and carbohydrate by weight, respectively, total energy contents by bomb calorimetry 4.2 kcal $\times$ g^{-1}, metabolizable energy 3.6 kcal $\times$ g^{-1}) or the same Chow to which had been added 50% by weight of lard. Nontested rats given the Purina Rat Chow during the whole experimental period followed as controls to evaluate the remaining effect of fasting during the refeeding period. Body weights, food consumption, and tissue weights (liver, muscle, and adipose tissues) were followed. Glycogen contents of liver and muscle, as well as incorporation of label from 1-^{14}C-glucose into carbondioxide and triglyceride in adipose tissue in vitro were also measured. Lipoprotein lipase was determined in adipose tissue according to Schotz et al (8) as modified by Hietanen and Greenwood (9).

Study II. Prevention of carbohydrate absorption

This study was also designed as described in detail previously (3, 4) but here during the refeeding phase a glucosidase inhibitor (Acarbose Bayer, Wuppertal, West Germany) at a concentration of 500 mg/kg Chow was added to the diet of one of the groups. Rats were fasted for 65 h and refed for 4 and 8 days.

Study III. Addition of sucrose during refeeding

In this study somewhat older male Sprague-Dawley rats were used (375 to 390 g). The fasting period was 65 h, and the refeeding period was 4 days. The feed used was from EWOS, Södertälje, Sweden containing 22.5, 55 and 5% of protein, carbohydrate, and fat by weight, respectively. One control group and one refed group were given an extra drinking bottle containing 10% sucrose, from which the rats drank ad libitum. Epididymal fat pads were removed after decapitation for determination of incorporation of 1-^{14}C-glucose into triglycerides in intact pieces of tissue as previously described (2) and for determination of lipoprotein lipase according to Nilsson-Ehle et al (10). Adipocyte size and weight were determined according to Sjöström et al (11) and triglyceride according to Carlson (12).

Statistical methods

"Student's" *t* test and conventional linear regression analyses were used. When more than two variables were compared, analysis of variance was applied.

Results

Study I

Figure 1 shows the body weight development in the nonfasted controls, the fasted rats refed with the ordinary Purina Rat Chow (carbohydrate rich) and the rats refed with fat-rich Purina Rat Chow. The rats refed the ordinary Chow were gaining weight faster than those refed fat-rich Chow.

The food intake is seen in **Figure 2**. Rats refed ordinary Chow started with a food intake below that of nonfasted controls, and then ate more during the rest of the period. The fat-fed animals show a caloric intake which was not different from nonfasted controls but less than animals refed ordinary Chow on day 3 and 4 ($p < 0.05$). With time the food intake in the fat-fed animals seemed to decline.

Weight of the tibialis anterior muscle was not different in fat refed rats as compared with rats refed ordinary Chow (**Fig 3**). Liver weight was lower in the fat-refed animals in comparison with Chow-refed animals on day 4 (Fig 3). The fat depots were, however,

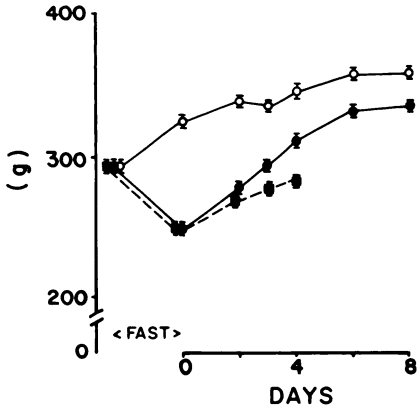

FIG 1. Body weight curves of rats fasted and refed ordinary Chow (*filled circles, solid line*) or fat-rich Chow (*filled squares, hatched line*) in comparison with nonfasted controls (*open circles, solid line*). Fat-refed animals lighter ($p < 0.05$) than Chow-refed animals at days 3 and 4. Means ± SEM (study I). (n = 8 in each group.)

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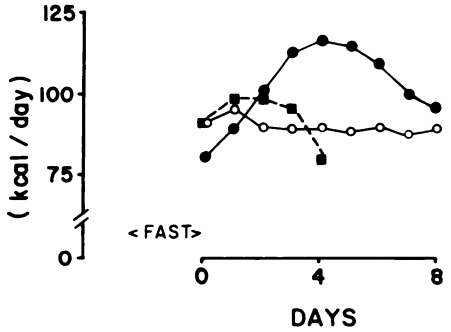

FIG 2. Food intake in rats fasted and refed ordinary Chow (*filled circles, solid line*) or fat-rich Chow (*filled squares, hatched line*) in comparison with nonfasted controls (*open circles, solid line*). To allow comparisons between weights of food intake of rats on fat-rich Chow and ordinary Chow the caloric density of the fat-rich Chow was corrected for by calculating values for utilizable calories (study I). (n = 8 in each group.)

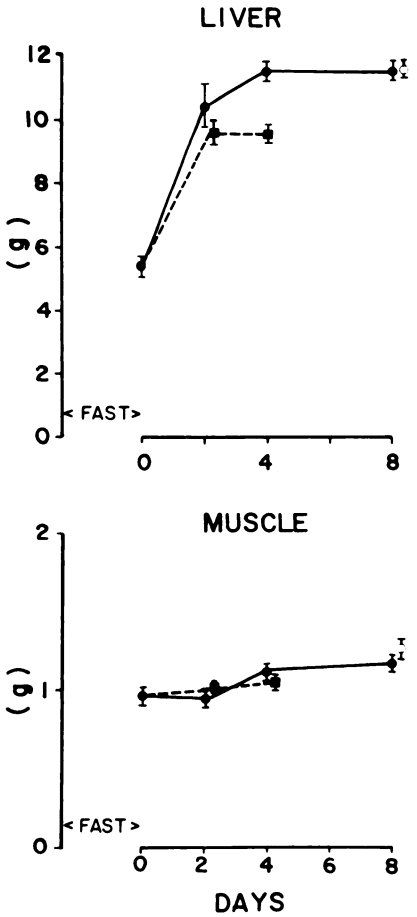

FIG 3. Weights of liver (*top*) and muscle (*bottom*, tibialis anterior) in rats fasted and refed ordinary Chow (*filled circles, solid lines*) or fat rich Chow (*filled squares, hatched line*) in comparison with nonfasted controls (*open circles*). Liver of fat-refed animals lighter than those of Chow-refed at day 4 ($p < 0.05$). Means ± SEM (study I). (n = 8 in each group.)

refilled faster in the fat-refed rats than in the rats refed ordinary Chow (**Fig 4**).

Unlike the animals refed the ordinary Chow, the rats refed fat Chow showed neither any overshoot in liver or muscle glycogen concentration during the first 4 days (**Fig 5**) nor in incorporation of label from glucose into carbon dioxide (**Fig 6**) or total triglyceride (**Fig 7**) in vitro in adipose tissue. Liver (day 2) and muscle (days 2 and 4) glycogen values in the fat-refed group were actually lower than those of nonfasted controls (Fig 5). **Table 1** shows that lipoprotein lipase activity was increased after refeeding, in comparison with nonfasted controls at all time points in both tissues ($p < 0.05$) (except perirenal fat pad refed Chow for 2 days). Rats refed fat showed higher values than those refed ordinary Chow at 2 days in the epididymal fat pad and at 4 days in the perirenal fat pad.

Study II

Table 2 shows that the body weight recovery showed a trend to be slower in the Acarbose group than in the group refed without acarbose at day 8. Liver weight was lower both at 4 and 8 days of refeeding in the Acarbose animals, probably corresponding to lower liver glycogen concentrations (significant at day 8, **Table 3**). Muscle weight did not differ with acarbose but glycogen contents were lower in Acarbose-fed rats at 4 days (Table 3). There was, however, a markedly diminished replenishment of fat depots, particularly in the retroperitoneal fat pad which had only about half the weight of this depot in animals refed Chow without Acarbose at 8 days (Table 2).

Food intake was not different in the groups with or without Acarbose (25 ± 2 and 24 ± 2 g/day during first 4 days, respectively, and 27 ± 3 g/day during all 8 days for both groups). Fecal contents of calories were 15.0 ± 2.6, 17.3 ± 1.9, and 23.6 ± 4.2% (means ± SEM) for the controls, the fasted-refed rats without or with Acarbose, respectively (percentage of ingested calories, measured with bomb calorimetry, calculated as averages

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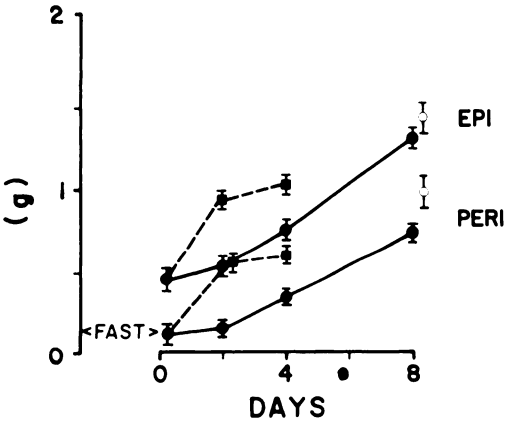

FIG 4. Weights of epididymal (*epi*) and perirenal (*peri*) fat pads in rats fasted and refed ordinary Chow (*filled circles, solid lines*) or fat-rich Chow (*filled squares, hatched lines*) in comparison with nonfasted controls (*open circles*). Adipose tissues of fat-refed animals heavier than those of Chow-refed at days 2 and 4 ($p < 0.05$). Means ± SEM (study I). (n = 8 in each group.)

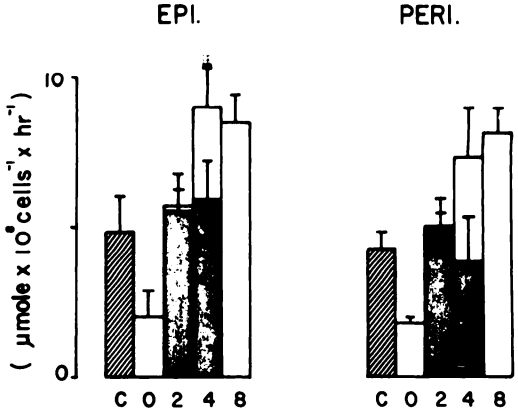

FIG 6. Incorporation of label from 1-^{14}C-glucose into carbon dioxide in rats fasted and refed ordinary Chow (*open bars*) or fat-rich Chow (*shadowed bars*) in comparison with nonfasted controls (*C, striped bars*). Significant ($p < 0.05$) differences in comparison with controls: *Epi.* and *Peri*: fasted (*lower*), 4 and 8 day refed Chow (*higher*).

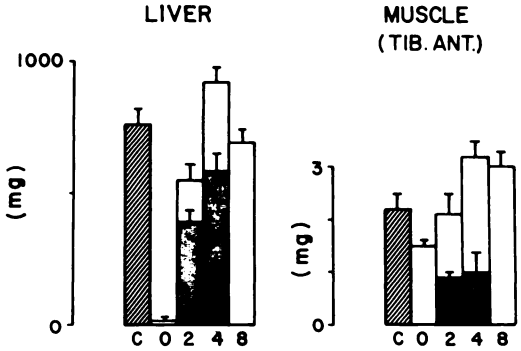

FIG 5. Glycogen contents of liver and muscle (tibialis anterior) in rats fasted and refed ordinary Chow (*open bars*) or fat-rich Chow (*shadowed bars*) in comparison with nonfasted controls (*C, striped bars*). Significant ($p < 0.05$) differences in comparison with controls: liver: fasted, 2-day refed fat or Chow (*lower*) 4 day refed Chow (*higher*). Muscle: fasted, 2 and 4 day refed fat (*lower*), 4 and 8 day refed Chow (*higher*). Means ± SEM (study I). (n = 8 in each group.)

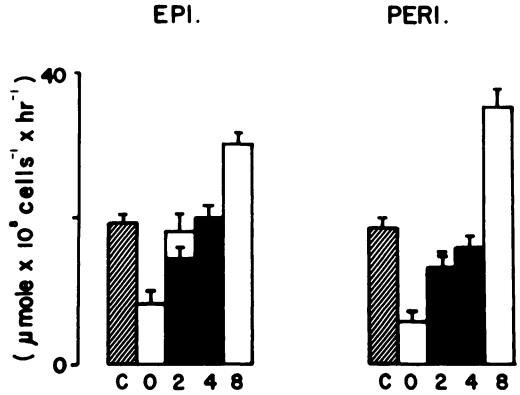

FIG 7. Incorporation of label from 1-^{14}C-glucose into triglyceride in rats fasted and refed ordinary Chow (*open bars*) or fat-rich Chow (*shadowed bars*) in comparison with nonfasted controls (*C, striped bars*). Significant ($p < 0.05$) differences in comparison with controls: *Epi*, and *Peri*: fasted (*lower*), 8 day refed Chow (*higher*).

over the 8 days of refeeding). These values are not significantly different. The nonutilizable energy of the Chow was 17%. None of the fecal contents of calories was different from this percentage.

Study III

Table 4 shows that the body weights of these fasted rats refed for 4 days as well as those in studies II and III were not back to control values. This was also the case with the weight of their epididymal fat pads and adipocytes. The sucrose consumed without restriction was equal in fasted-refed rats or control rats. Sucrose consumption caused larger fat cells in the ad libitum fed (control) rats ($p < 0.05$).

Both incorporation of glucose-label into triglycerides and lipoprotein lipase activity was dependent on fat cell size. When compared at the same fat cell size, fasting-refeeding caused an increase in both activities. Sucrose addition had, however, no effect on either activity (not shown).

Table 5 shows the effects of fasting-refeeding on the insulin response of triglyceride

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TABLE 1
Lipoprotein lipase activity in adipose tissues (study 1)

	n	Epididymal fat pad	Perirenal fat pad
Controls	8	1.20 ± 0.35	0.92 ± 0.15
Fasted	6	0.39 ± 0.05	0.39 ± 0.08
Refed Chow, 2 days	6	3.19 ± 0.37	1.15 ± 0.27
Refed fat, 2 days	6	4.82 ± 0.34*	2.91 ± 1.21
Refed Chow, 4 days	6	3.77 ± 0.19	1.47 ± 0.20
Refed fat, 4 days	6	2.88 ± 0.50	3.43 ± 0.76*
Refed Chow, 8 days	6	3.62 ± 0.20	2.07 ± 0.23

* $p < 0.05$ compared with Chow fed, μmol free fatty acid × mg protein^{-1} × h^{-1}, means ± SEM.

TABLE 2
Body and organs weight of rats refed with or without Acarbose (study II)

	n	Body wt	Liver wt	Muscle* wt	Epididymal fat pad wt	Perirenal fat pad wt
		g	g	g	g	g
Controls	7	333 ± 9	12.9 ± 0.5	0.57 ± 0.02	2.21 ± 0.21	1.86 ± 0.24
Fasted	6	167 ± 4	4.4 ± 0.2	0.32 ± 0.01	0.52 ± 0.14	0.21 ± 0.13
Refed 4 days	7	244 ± 4	10.4 ± 0.2	0.41 ± 0.01	0.94 ± 0.08	0.52 ± 0.04
Refed 4 days with Acarbose	7	247 ± 4	9.5 ± 0.2†	0.41 ± 0.01	0.83 ± 0.06	0.35 ± 0.04†
Refed 8 days	7	294 ± 9	11.8 ± 0.4	0.51 ± 0.20	1.73 ± 0.11	1.29 ± 0.21
Refed 8 days with Acarbose	7	272 ± 5‡	10.7 ± 0.2†	0.47 ± 0.11	1.26 ± 0.06§	0.66 ± 0.04§

* M tibialis anterior. Comparisons between groups refed with or without Acarbose.
† $p < 0.05$.
‡ $p < 0.10 > 0.05$.
§ $p < 0.01$. Means ± SEM.

TABLE 3
Liver and muscle glycogen contents in rats refed with or without Acarbose (study II)

	n	Liver glycogen (mg per whole tissue)	Muscle* glycogen (mg per whole tissue)
Controls	7	409 ± 48	1.64 ± 0.42
Fasted	6	12 ± 6	0.32 ± 0.08
Refed 4 days	7	485 ± 43	1.93 ± 0.09
Refed 4 days with Acarbose	7	376 ± 40	1.55 ± 0.16†
Refed 8 days	7	503 ± 39	1.70 ± 0.16
Refed 8 days with Acarbose	7	377 ± 47†	1.75 ± 0.13

* M tibialis anterior. Comparisons between groups refed with or without Acarbose.
† $p < 0.05$. Means ± SEM.

TABLE 4
Effects of sucrose addition during refeeding on body, epididymal fat pad and adipocyte weights (study III)

	Ad libitum fed controls	Fasted-refed	Ad libitum fed controls (plus sucrose)	Fasted-refed (plus sucrose)
Body wt at start (g)	392 ± 4	384 ± 1	395 ± 2	388 ± 3
Body wt after starving period (g)	412 ± 4	281 ± 2	412 ± 3	285 ± 3*
Body wt after refeeding period (g)	411 ± 3	323 ± 2*	413 ± 3	325 ± 4*
Wt of epididymal fat pads after refeeding period (g)	4.71 ± 0.13	1.94 ± 0.15*	4.81 ± 0.25	2.19 ± 0.13*
Adipocyte wt after refeeding period (μg)	0.185 ± 0.008	0.049 ± 0.004*	0.212 ± 0.012	0.071 ± 0.013*
Sucrose consumed (g)	0	0	9.9 ± 1.8	9.8 ± 0.5

* $p < 0.001$ in comparison with respective controls. Means ± SEM. n = 10 in all groups.

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TABLE 5
Effects of sucrose addition during refeeding on basal and insulin stimulated incorporation of 1-^{14}C-glucose into triglycerides (study III)

	Ad libitum fed controls	Fasted-refed	Ad libitum fed controls (plus sucrose)	Fasted-refed (plus sucrose)
Basal	78 ± 5	73 ± 12	95 ± 6	96 ± 17
Increase by insulin (100 μU/ml)	33 ± 4	47 ± 15	41 ± 8	75 ± 12*
Relative increase (%)	42 ± 4	73 ± 21	45 ± 9	88 ± 15*
Increase by insulin (1000 μU/ml)	50 ± 8	57 ± 15	58 ± 12	123 ± 28*
Relative increase (%)	63 ± 8	88 ± 18	63 ± 12	135 ± 24†

* $p < 0.05$. † $p < 0.01$ in comparisons with ad libitum fed controls (plus sucrose). The same differences are found in comparisons between fasted-refed (plus sucrose) rats and fasted-refed rats except the relative increase (%) by 100 μU/ml insulin.

Values given are μmol glucose/10^6 cells/hr, or where indicated, percentage. Means ± SEM. n = 10 in all groups.

synthesis from glucose in adipose tissue. Although fasted-refed rats not given sucrose showed higher absolute and relative average responses to insulin, none of these was significantly different from corresponding values of controls fed ad libitum. When sucrose was added to the refeeding diet the basal level of triglyceride label was not different from that of rats not given sucrose. The insulin response was, however, increased at both insulin concentrations tested.

Discussion

The rationale for the selection of these three experimental designs was the following. Previous experiments have indicated the importance of carbohydrate for the development of several of the adaptations occurring on refeeding after fasting (2–4). It was therefore natural to include experiments where less carbohydrate was provided in the Chow (study I) or where the absorption of carbohydrate in the Chow was slowed (study II).

In study III specific effects of carbohydrate ingestion were examined in the main tissue for energy storage, viz, adipose tissue, in its main pathways for lipid accumulation, viz, triglyceride synthesis from glucose and lipoprotein lipase activity, responsible for regulation of storage of exogenous fat.

Study I shows that decreasing carbohydrate intake during refeeding diminishes several adaptations in tissue energy storage activities, including liver weight and glycogen contents as well as lipid synthesis in adipose tissue. Indeed, glycogen concentrations in liver and muscle were not even back to control values, presumably due to lack of carbohydrate substrate. On the other hand, pathways for assimilation of exogenous lipid are adapted instead, lipoprotein lipase activity was elevated, and lipid accumulation in the adipose stores was very effective at least during the first 4 days of refeeding. This then indicates that the adaptive phenomena with refeeding after fasting are not exclusive for the pathways of carbohydrate storage, they occur also effectively in the pathways for lipid storage.

This question was tested in more detail in study III where specific effects of added dietary sucrose were studied on the main pathways for carbohydrate and lipid storage in adipose tissue. Here sucrose had specific effects on insulin responsiveness of adipocytes, but no effect on lipoprotein lipase activity.

Study II examined the effects on adaptive phenomena of interference with carbohydrate absorption by a glucosidase inhibitor. The stool losses of energy here were not different between groups, and not different from that expected from the non-utilizable contents of the Chow. This indicates that the glucosidase inhibitor does not cause carbohydrate malabsorption. It seems more likely that it causes a delay in carbohydrate absorption. In these experiments liver glycogen repletion was delayed, and more strikingly, the refilling of the fat depot was markedly inhibited. This might well have been due to less activation of lipoprotein lipase in these adipose tissues. Such an activation is dependent on glucose metabolism via insulin action (13),

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which might have been less pronounced due to delayed carbohydrate absorption.

During the period studied there was no difference in energy intake or stool losses in the group given the glucosidase inhibitor as compared with controls. Nevertheless, these animals showed less rapid energy reaccumulation. These findings suggest that this drug causes a decreased food efficiency, possibly by increasing thermogenetic mechanisms.

The adaptive responses to refeeding after fasting in the animal model used in the present and previous works might well be of significance for the difficulties of treated human obese subjects to avoid relapse. Refeeding weight-reduced obese subjects with fat is not a good alternative as judged from these experiments here, body weight and carbohydrate stores are reaccumulated less efficiently, but the lipid stores are quickly refilled, a consequence that clearly is not desirable. Inhibition or delay of carbohydrate absorption may, however, be a better alternative, because this seems to prevent a rapid refilling of the fat stores. The reason for this might be that glucose is less rapidly absorbed due to the action of this drug, and may be due to less pronounced adaptations of gut enzymes after refeeding (5). This might have as a consequence both less storage of liver glycogen, and less storage of fat in the depots. An effect on thermogenesis might also be considered.

Clinical trials to treat human obesity with the glucosidase inhibitor used here have not been promising, weight decreases amounting to no or insignificant losses of weight (William-Olsson T, Sjöström L, unpublished data). It might well be, however, that this principle of treatment should be tried along lines suggested by the results obtained in the present work, viz, to prevent or delay relapse after conventional dietary treatment. Such a study is currently being performed (William-Olsson T, Sjöström L, unpublished data).

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